Steering equations for deception probes

Deception-probes project

Symbols

Symbol	Meaning
$\boldsymbol{h} \in \mathbb{R}^d$	residual-stream (hidden-state) vector for one example, read at probe layer L; d is the model's hidden size
$\boldsymbol{h}'$	the same vector after the steering intervention (what generation actually uses)
d	hidden dimension (residual-stream width) of the model
L	probe layer: the index of the residual stream the probe reads / we steer
$\boldsymbol{v}$	probe direction, a *unit* vector ($\lVert\boldsymbol{v}\rVert = 1$); the axis we add or remove
b	probe bias (scalar): the calibrated decision threshold in $\text{score}(\boldsymbol{h}) = \boldsymbol{h} \cdot \boldsymbol{v} - b$
c	a class: caved (1, deceptive) or held (0, control); $\lvert c\rvert$ is its example count
$\boldsymbol{\mu}_c$	mean activation vector of class c
$\boldsymbol{w}$	logistic-regression weight vector, learned in standardized space
$\boldsymbol{\sigma}$	per-dimension standard deviation of activations (the standardizer); $\oslash$ = elementwise divide
α	steering strength (dimensionless coefficient), swept on the x-axis of the add-pass plots
s	layer scale: mean last-token residual norm at layer L; makes α comparable across layers/models
$\boldsymbol{r}$	matched random unit vector ($\lVert\boldsymbol{r}\rVert = 1$) used as the control direction
N_{parsed}	number of items whose steered generation produced a parseable answer
$\mathbb{1}[\cdot]$	indicator: 1 if the condition holds, else 0
I	the $d \times d$ identity matrix
$\boldsymbol{v}\boldsymbol{v}^\top$	outer product ($d \times d$ rank-1 matrix); $I - \boldsymbol{v}\boldsymbol{v}^\top$ projects onto the complement of $\boldsymbol{v}$

Class-label convention (never flipped): 1 = deceptive (caved / sandbagged), 0 = control (held).

1. Probe direction

Difference-of-means (`mms`) — the estimator that steers:

$$\boldsymbol{v}_{\text{mms}} = \frac{\boldsymbol{\mu}_{\text{caved}} - \boldsymbol{\mu}_{\text{held}}}{\lVert\boldsymbol{\mu}_{\text{caved}} - \boldsymbol{\mu}_{\text{held}}\rVert}, \qquad \boldsymbol{\mu}_c = \frac{1}{\lvert c\rvert} \sum_{\boldsymbol{h} \in c} \boldsymbol{h}. \tag{1}$$

Logistic regression (`lr`), fit on standardized activations with weight $\boldsymbol{w}$ and per-dimension standard deviation $\boldsymbol{\sigma}$, then folded back to raw space:

$$\boldsymbol{v}_{\mathrm{lr}} = \frac{\boldsymbol{w} \oslash \boldsymbol{\sigma}}{\|\boldsymbol{w} \oslash \boldsymbol{\sigma}\|}, \qquad (\oslash = \text{elementwise division}). \tag{2}$$

Scoring an activation (projection minus a calibrated bias b):

$$\operatorname{score}(\boldsymbol{h}) = \boldsymbol{h} \cdot \boldsymbol{v} - b. \tag{3}$$

2. Alpha and the layer scale

The steering coefficient α is dimensionless; the injected magnitude is α times the layer's mean last-token residual norm s:

$$s = \operatorname*{mean}_i \|\boldsymbol{h}_i\| \quad (\text{layer } L,\ \text{last token}), \qquad \text{injected vector} = \alpha\, s\, \boldsymbol{v}. \tag{4}$$

With the `--raw` flag the scale is dropped: injected vector $= \alpha\, \boldsymbol{v}$.

3. Add pass (sufficiency)

On unpressured items answered correctly at baseline, add the direction at layer L, then generate:

$$\boldsymbol{h}' = \boldsymbol{h} + \alpha\, s\, \boldsymbol{v}. \tag{5}$$

Causal signature: the wrong-answer rate rises with α, over parsed items,

$$\text{wrong\_rate}(\alpha) = \frac{1}{N_{\text{parsed}}} \sum_{i=1}^{N_{\text{parsed}}} \nVdash[\,\text{answer}_i(\alpha) \text{ is wrong}\,]\,. \tag{6}$$

4. Ablate pass (necessity)

On pressured items that cave at baseline, project the direction out, then generate:

$$\boldsymbol{h}' = \boldsymbol{h} - (\boldsymbol{h} \cdot \boldsymbol{v})\, \boldsymbol{v} = \left(I - \boldsymbol{v}\boldsymbol{v}^\top\right) \boldsymbol{h}. \tag{7}$$

Here $P = I - \boldsymbol{v}\boldsymbol{v}^\top$ is the orthogonal projector onto the complement of $\boldsymbol{v}$ (valid because $\|\boldsymbol{v}\| = 1$). Causal signature: the caving rate drops versus baseline.

5. Matched random control

Each pass is repeated with a random unit vector $\boldsymbol{r}$, $\|\boldsymbol{r}\| = 1$, in place of $\boldsymbol{v}$:

$$\boldsymbol{h}' = \boldsymbol{h} + \alpha\, s\, \boldsymbol{r} \qquad (\text{add}), \qquad \boldsymbol{h}' = \boldsymbol{h} - (\boldsymbol{h} \cdot \boldsymbol{r})\, \boldsymbol{r} \qquad (\text{ablate}). \tag{8}$$

A direction counts as causal only if its curve clears the random curve, ruling out generic perturbation.

6. Layer indexing

The probe reads hidden_states[L], the output of decoder block L (index 0 is the embedding layer). The intervention hook therefore attaches to layers[$L-1$], whose output *is* hidden_states[L]. Layer 0 cannot be steered (no block produces it).

Instructed-arm example values (Qwen2.5-7B, mms, L=20, prompt-final): $s = 89.3$; add pass wrong_rate $= 0 \to 0.39 \to 0.46 \to 0.72$ *at* $\alpha = 0, 0.25, 0.5, 1.0$ *(random* ≈ 0.013*); ablate* $1.00 \to 0.835$ *(random* 0.985*).*